

Business Requirements Document (BRD)

Project Metadata

- **Project Title:** AdventureWorks Enterprise Sales Analytics Platform
- **Document Version:** 1.0
- **Domain:** Sales & Commercial Analytics
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- **Status:** Draft for Review

1. Executive Summary

AdventureWorks is a global bicycle manufacturing and multi-channel retail company specializing in high-quality cycling equipment, components, clothing, and accessories. The business operates a dual-channel distribution model: direct-to-consumer online sales (**Internet Sales**) and business-to-business retail partnerships (**Reseller Sales**).

Currently, leadership and operational teams suffer from fragmented data sources, hindering their ability to monitor comprehensive performance, evaluate regional growth, measure individual sales targets, and track channel profitability.

This project delivers a centralized, enterprise-grade Sales Analytics Platform built on top of the **AdventureWorksDW** data warehouse. By utilizing robust semantic modeling (Power BI/Tabular) and industry-standard metrics, this platform will serve as the single source of truth for strategic, tactical, and operational sales decisions across the enterprise.

2. Business Background

AdventureWorks structures its commercial operations across two distinct primary sales channels, supplemented by an internal sales performance management framework:

A. Internet Sales (Direct-to-Consumer / B2C)

Products are sold directly to global end-users through the company's digital e-commerce platform.

- **Product Portfolio:** Complete Bikes, Components, Technical Clothing, and Accessories.
- **Customer Profile:** Individual consumers with direct, transactional relationships.

B. Reseller Sales (Business-to-Business / B2B)

Products are distributed to specialized value-added resellers, local bicycle stores, and multinational sporting goods retailers.

- **Product Portfolio:** Bulk shipments of complete bikes, frames, and commercial

components.

- **Customer Profile:** Corporate accounts, distributors, and physical retail outlets.

C. Sales Performance Management

The organization employs dedicated sales representatives assigned to specific territories. To drive revenue and optimize performance, the corporate office establishes periodic sales quotas (targets) for individual employees and geographic regions. Management requires structured visibility into target achievements to align incentives and strategic expansions.

3. Business Problem Statement

The primary challenge facing AdventureWorks is the **lack of a unified, consolidated analytical layer** capable of aggregating sales performance across both B2C (Internet) and B2B (Reseller) channels.

Currently, business users and decision-makers face severe bottlenecks in the following areas:

- **Fragmented Revenue Tracking:** Difficulty in viewing consolidated corporate revenue, leading to delayed financial health assessments.
- **Inability to Compare Channels:** No standardized method to contrast performance, market share, and operational efficiency between Internet and Reseller channels.
- **Opaque Profitability:** Margin calculations are inconsistent across departments, making it hard to identify low-margin segments or product lines.
- **Siloed Target Evaluation:** Tracking sales representatives' quota achievements requires manual, spreadsheet-based consolidation, leading to delayed performance feedback.
- **Underutilized Customer & Demographic Data:** Inability to effectively segment high-value customers or analyze geographic purchasing trends.

Ultimately, these limitations slow down decision-making, increase reliance on manual reporting, and limit the company's ability to act on market opportunities in real-time.

4. Strategic Business Objectives

The analytics platform is designed to achieve the following core objectives:

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| **Objective ID** | **Objective Title** | **Description** |

| **BO-01** | **Unified Sales Visibility** | Establish a centralized analytical platform that merges Internet and Reseller sales channels to provide a holistic view of enterprise performance. |

| **BO-02** | **Standardized Financial Metrics** | Deliver unified definitions and computations for core revenue, cost, profit, and margin indicators across all channels. |

| **BO-03** | **Product & Customer Intelligence** | Identify high-value customer segments and

discover top-performing product categories, subcategories, and individual SKUs. |

| **BO-04 | Performance & Quota Tracking** | Provide dynamic monitoring of sales targets (quotas) versus actual achievements for sales representatives and territories. |

| **BO-05 | Data-Driven Culture** | Empower regional managers and executive leadership with self-service, interactive BI reports to accelerate operational decisions. |

5. Scope of Work

5.1 In-Scope (Version 1.0)

The initial release of the sales analytics platform will ingest data from the following tables within AdventureWorksDW and focus on specific analytical areas:

A. Data Layer & Schema Elements

- **Fact Tables:**
 - FactInternetSales (Online transactional sales)
 - FactResellerSales (B2B distributor sales)
 - FactSalesQuota (Representative sales targets)
- **Dimension Tables:**
 - DimDate (Standard and fiscal calendars)
 - DimProduct, DimProductSubcategory, DimProductCategory (Product hierarchy)
 - DimCustomer (B2C demographic profiles)
 - DimReseller (B2B client profiles)
 - DimEmployee (Sales hierarchy and representatives)
 - DimPromotion (Marketing and discount actions)
 - DimCurrency (Multi-currency transaction mapping)
 - DimSalesTerritory (Sales organizational geography)
 - DimGeography (Regional and spatial data)

B. Core Analytical Areas

- **Revenue & Growth Dynamics:** Historical trend analysis, seasonal performance, and period-over-period growth.
- **Profitability Analytics:** Cost of Goods Sold (COGS) vs. Gross Profit Margin comparison across regions and products.
- **Customer Demographics & Segmentation:** Buying behavior, purchase frequency, and spatial distribution of B2C buyers.
- **Product Catalog Performance:** Classifying products into high, average, and underperforming categories.
- **Territory & Spatial Analysis:** Revenue mapping based on regional hierarchies.
- **Representative Quota Achievement:** Direct comparison of actual sales metrics against set quotas.
- **Dual-Channel (B2C vs. B2B) Performance:** Comparative views of volume, margins, and

transaction sizes.

5.2 Out-of-Scope (Version 1.0)

The following domains are explicitly excluded from the current phase to ensure a focused, high-quality, and timely delivery. They are earmarked for future iterations:

- **Inventory & Warehousing Analytics:** (e.g., Stock status, safety margins, turnover rates).
- **Financial & Corporate Ledger Reporting:** (e.g., Balance Sheets, Cash Flow, tax compliance).
- **Manufacturing & Supply Chain Analytics:** (e.g., Bill of Materials, factory downtime, scrap rates).
- **Procurement & Vendor Management:** (e.g., Purchase orders, vendor lead times, vendor pricing).
- **HR & Workforce Management:** (e.g., Employee attrition, payroll, non-sales employee performance).
- **Call Center & Customer Support Analytics:** (e.g., Ticket counts, resolution times, CSAT).

6. Stakeholders & Key Users

| **Stakeholder Persona** | **Key Responsibility in Organization** | **Platform Use Case & Value Proposition** |

| **Executive Management** | Strategic performance, corporate direction, board reporting. | High-level KPI dashboards focusing on overall revenue, global margins, and YoY growth. |

| **Sales Director** | Sales strategy execution, channel health, quota setting. | High-level overview of channel performance (B2B vs B2C) and regional target achievement rates. |

| **Sales Managers** | Team productivity, regional operations, representative coaching. | Detailed reports on individual sales representative performance against targets and territory growth. |

| **Product Managers** | Portfolio management, inventory planning, pricing adjustments. | Deep-dive reports on product categories, slow-moving items, and promotional performance. |

| **Business & Data Analysts** | Custom analysis, ad-hoc reports, regional deep-dives. | Standardized semantic model access via Power BI to build custom, ad-hoc dashboards. |

7. Key Business Questions (KBQs)

The platform must provide immediate, clear answers to the following operational and strategic questions:

A. Revenue & Growth

- **BQ-01:** What is the total enterprise revenue generated across all channels?
- **BQ-02:** How is revenue trending over days, months, quarters, and fiscal years?

- **BQ-03:** What are the period-over-period (MoM, YoY) growth rates for revenue and volume?

B. Product Performance

- **BQ-04:** Which specific products contribute the highest absolute sales and net profits?
- **BQ-05:** Which product categories and subcategories are generating the strongest margins?
- **BQ-06:** Which products are underperforming or showing declining sales velocity?

C. Customer Behavior

- **BQ-07:** Who are our highest-value customers (B2C and B2B) based on lifetime spending?
- **BQ-08:** Which customer demographic segments (e.g., gender, income, geographic location) contribute most to online sales?
- **BQ-09:** How is customer purchasing behavior evolving over time (average order value, unit counts)?

D. Territory Analytics

- **BQ-10:** Which global territories and regions generate the highest sales volume?
- **BQ-11:** Which regional markets are expanding the fastest, and which are stagnating?

E. Channel Comparison

- **BQ-12:** What is the sales and volume distribution between Internet Sales and Reseller Sales?
- **BQ-13:** Which channel yields the higher Gross Profit Margin, and how do their average order sizes compare?

F. Quota & Employee Performance

- **BQ-14:** Are global and regional sales targets (quotas) currently being achieved?
- **BQ-15:** Which sales representatives are consistently exceeding their set quotas, and who are the top performers?
- **BQ-16:** Which specific territories are falling short of their targets, requiring operational intervention?

8. Key Performance Indicators (KPIs) & Formulations

The following core business metrics are standardized for the AdventureWorks Sales Analytics Platform. All calculations must be implemented precisely in the semantic model:

8.1 Total Revenue

- **Definition:** Total gross sales amount generated from transactions before deductions.
- **Mathematical Formula:**

$$\text{Revenue} = \sum(\text{SalesAmount})$$

8.2 Total Product Cost

- **Definition:** Total cost of goods sold (COGS), representing the direct manufacturing or acquisition cost of the products sold.
- **Mathematical Formula:**

$$\text{Cost} = \sum(\text{TotalProductCost})$$

8.3 Gross Profit

- **Definition:** The absolute profit earned after subtracting the Cost of Goods Sold from total sales revenue.
- **Mathematical Formula:**

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

8.4 Gross Profit Margin Percentage

- **Definition:** A metric representing profitability as a percentage of total sales revenue.
- **Mathematical Formula:**

$$\text{Margin \%} = \left(\frac{\text{Profit}}{\text{Revenue}} \right) \times 100$$

8.5 Sales Quantity

- **Definition:** The total cumulative units of products sold.
- **Mathematical Formula:**

$$\text{Sales Quantity} = \sum(\text{OrderQuantity})$$

8.6 Quota Achievement Percentage

- **Definition:** The percentage of actual sales generated by an employee or territory relative to their established sales target.
- **Mathematical Formula:**

$$\text{Quota Achievement \%} = \left(\frac{\text{Actual Sales}}{\text{Sales Quota}} \right) \times 100$$

- *Note: Actual Sales here is mapped from FactResellerSales.SalesAmount for representatives.*

8.7 Year-over-Year (YoY) Growth Percentage

- **Definition:** The relative change in sales revenue for a specific period compared to the exact same period in the previous calendar/fiscal year.
- **Mathematical Formula:**

$$\text{YoY Growth \%} = \left(\frac{\text{Revenue}_{\text{Current Period}} - \text{Revenue}_{\text{Prior Year Period}}}{\text{Revenue}_{\text{Prior Year Period}}} \right) \times 100$$

8.8 Month-over-Month (MoM) Growth Percentage

- **Definition:** The relative change in sales revenue for a specific month compared to the preceding month.
- **Mathematical Formula:**

$$\text{MoM Growth \%} = \left(\frac{\text{Revenue}_{\text{Current Month}} - \text{Revenue}_{\text{Previous Month}}}{\text{Revenue}_{\text{Previous Month}}} \right) \times 100$$

9. Reporting & Analytical Experiences

To accommodate varying analytical needs, the Power BI solution will provide three distinct reporting interfaces:

9.1 Executive Performance Dashboard

- **Target Audience:** C-Suite Executives, Regional Directors, and the Sales VP.
- **Analytical Focus:**
 - Consolidated global KPIs (Revenue, Profit, Margins, Sales Units) with clear YoY indicators.
 - Geographical sales distribution via interactive map visualizations.
 - Trend lines highlighting revenue trajectories across fiscal years.
 - High-level channel division comparison.

9.2 Tactical Sales Operations & Quota Tracker

- **Target Audience:** Sales Directors, HR Managers, and Sales Territory Managers.

- **Analytical Focus:**
 - Interactive grid comparison of actual sales vs. quota targets by employee.
 - Territory performance breakdowns to identify lagging or leading geographical teams.

9.3 Product & Customer Intelligence Module

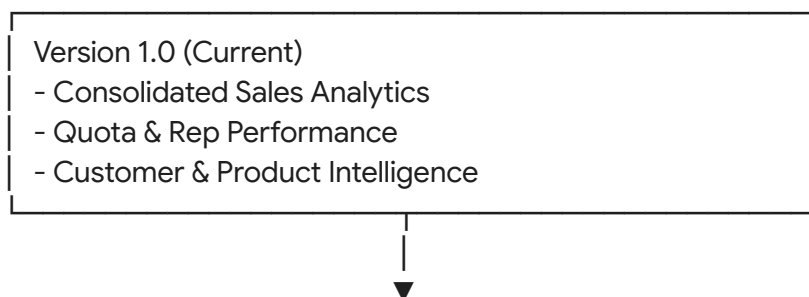
- **Target Audience:** Product Category Managers, Inventory Planners, and Marketing Specialists.
- **Analytical Focus:**
 - Matrix views highlighting sales, quantities, and profit margins down to the individual SKU level.
 - Pareto analysis (80/20 rule) identifying top-tier customers and high-volume products.
 - Customer demographic slicing (Income bracket, Occupation, Region) to optimize targeted marketing campaigns.

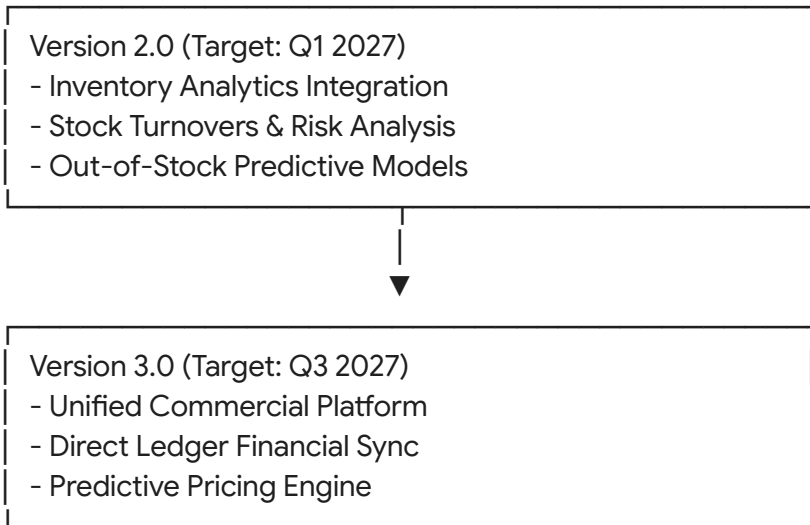
10. Success Criteria

The implementation of the AdventureWorks Sales Analytics Platform will be deemed successful when the following milestones are met and verified:

- **SC-01 (Cross-Channel Integration):** Business stakeholders can review and contrast transaction-level metrics for both Internet and Reseller sales channels inside a single semantic model.
- **SC-02 (Metric Standardization):** Every department utilizes identical, verified DAX measures for calculating Gross Profit, Margin %, and period-over-period growth, eliminating reporting discrepancies.
- **SC-03 (Automated Target Monitoring):** Manual Excel-based quota matching is eliminated. The system dynamically updates quota achievement percentages as new daily sales data flows in.
- **SC-04 (Actionable Insights):** Product and Category Managers can identify underperforming products and high-value customer segments within three clicks.
- **SC-05 (Strategic Enablement):** Regional performance dashboards are integrated into monthly executive review meetings to support faster, data-backed decisions.

11. Future Roadmap





Phase 2: Inventory Analytics Data Product (Version 2.0)

- Integration of the FactProductInventory table.
- Implementation of metrics tracking Inventory Turnover Ratio, Average Days in Stock, and Stock Availability percentages.
- Automated alert generation for products showing a high risk of stockouts or excessive warehouse holding costs.

Phase 3: Unified Commercial Analytics Platform (Version 3.0)

- Complete convergence of sales performance, active supply chain statuses, and raw procurement ledger tables.
- Predictive modeling to forecast future sales trends based on historical seasonal data, promotional campaigns, and regional economic indicators.